φ

Phi

A constructed language

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Table of Contents

1 Introduction

PHI is a constructed language. A constructed language is one whose phonology, grammar, and vocabulary, instead of having developed naturally, are consciously devised.[1] The sound of the name *Phi* embodies the simplicity and softness of the language itself. *Phi (Φ)* is also beauty in mathematics. It is

> *the golden ratio, an irrational number with a value of approximately 1.618033988 which expresses the relationship that the sum of two quantities is to the larger quantity as the larger is to the smaller.* – Wiktionary[2]

It is thought that the value of *phi* is one of the fundamental characteristics of nature; in the language Phi, nature is fundamental to its being. The natural world is the root of many of the words in the Phi lexicon. More industrial or unnatural words are constructed by describing the subject in terms of existing, more natural words. As an historical (or perhaps Hollywood) example, native North Americans called a rifle a *fire stick*. They described something in terms with which they were familiar.

There is also an emphasis on simplicity in the language. Phi has many aspects of a pidgin language without having been derived from existing languages. An a priori pidgin is definitely a contradiction in terms, but the resemblance to a pidgin is intentional. Wikipedia describes some of the common traits of such a language.[3] These are the similarities in Phi to those traits:

- Phi is a isolating language but not an extreme example of one;
- It is also an analytic language, but taken further than most pidgins;
- There are no embedded clauses in Phi;
- Phi eschews syllable codas;
- Consonant clusters are restricted to voiceless digraphs, e.g. /sh/;
- Aspiration only occurs with initial /p/ and /t/;
- Phi uses the common five-vowel system: /i/, /u/, /e/, /o/, /a/;
- There is no sound influence between morphemes;

[1] `https://en.wikipedia.org/wiki/Constructed_language`
[2] `https://en.wiktionary.org/wiki/%CF%86`
[3] `https://en.wikipedia.org/wiki/Pidgin#Common_traits`

- There are no tones in Phi;
- Phi has no grammatical tense;
- No conjugation or declension occurs;
- There is a lack of grammatical gender or number.

These traits are where Phi diverges from a pidgin:

- There are no historical sound changes in an a priori language;
- No monopthongization occurs in Phi: each vowel is pronounced separately;
- Phi has clear parts of speech and word categorization;
- In addition, Phi uses a SOV word order.

Phi is an attempt to answer a few questions. What if humans evolved to cooperate with nature instead of trying to dominate it? What if language remained simple and straightforward instead of growing in complexity as society advanced? What if society naturally tended toward the positive instead of the negative? What would that language look and sound like? In this respect Phi can be seen as an engineered language, specifically a philosophical language.[4]

[4] https://en.wikipedia.org/wiki/Philosophical_language

2 Phonology

2.1 Consonants

Table 1: Consonant phonology

	Bilabial	Labial	Dental	Alveolar	Postalv.	Velar	Glottal
Plosive	p		t̪				
Nasal	m		n̪				
Trill				r			
Tap or flap				ɾ			
Fricative	ɸ	ʍ	θ	s	ʃ	ʍ	h
Approximate		w				w	
Lateral approx.				l			

2.2 Vowels

Table 2: Vowel phonology

	Front	Central	Back
Close	i		u
Mid	e̞		o̞
Open		ä	

3 Orthography

3.1 Consonants

Table 3: Consonant orthography

Allophone	Grapheme	Description
[p]	‹p›	Voiceless bilabial plosive
[t̪]	‹t›	Voiceless laminal dental plosive
[m]	‹m›	Voiced bilabial nasal
[n̪]	‹n›	Voiced laminal dental nasal
[r]	‹r›	Voiced alveolar trill
[ɾ]	‹r›	Voiced alveolar tap or flap
[ɸ]	‹ph›	Voiceless bilabial fricative
[ʍ]	‹wh›	Voiceless labialized velar approximant
[θ]	‹th›	Voiceless dental fricative
[s]	‹s›	Voiceless alveolar fricative
[ʃ]	‹sh›	Voiceless postalveolar fricative
[h]	‹h›	Voiceless glottal fricative
[w]	‹w›	Voiced labialized velar approximant
[l]	‹l›	Voiced alveolar lateral approximant

3.2 Vowels

Table 4: *Vowel orthography*

Allophone	Grapheme	Description
[i]	‹i›	Close front unrounded
[u]	‹u›	Close back rounded
[e̞]	‹e›	Mid front unrounded
[o̞]	‹o›	Mid back rounded
[ä]	‹a›	Open central unrounded

3.3 Vowel Pairs

Vowel pairs are never diphthongs in Phi; they are always pronounced separately. The vowel pairs specified in Table 5 show the vowels in hiatus.

Table 5: *Vowel pair orthography*

Allophone	Grapheme
[i.e̞]	‹ie›
[i.o̞]	‹io›
[i.ä]	‹ia›
[e̞.i]	‹ei›
[e̞.o]	‹eo›
[e̞.ä]	‹ea›
[o̞.i]	‹oi›
[o̞.e̞]	‹oe›
[o̞.ä]	‹oa›
[ä.i]	‹ai›
[ä.e̞]	‹ae›
[ä.o̞]	‹ao›

4 The Tengwar

Phi can be written using the standard Latin alphabet, but is usually written using J. R. R. Tolkien's Tengwar[5] script. The Tengwar was first described in Appendix E of the third book in Tolkein's trilogy, *The Lord of the Rings*. There are numerous sources of information available concerning the Tengwar, so the history and details of the script will not be explored in this document.

The look and feel of the Tengwar's characters (*tengwa*) and its diacritical marks (*tehta*) match the ideals and philosophies of this language. Phi has a small set of consonants and vowels so it is relatively straightforward to map its graphemes to Tengwar characters. The Unicode font *Tengwar Telcontar*[6] is used in this manual to render Phi text in Tengwar script. A custom keyboard layout[7] was employed to facilitate typing Tengwar characters.

Tables 6 and 7 outline the mapping between the graphemes used in Phi and the corresponding Tengwar characters, as well as the keypresses necessary to produce the characters using the font and the keyboard layout mentioned previously. An application or utility that supports Graphite fonts[8] is required for the font to render correctly. A link to details of this requirement can be found in the second footnote below.

Phi uses the Tengwar script in its own mode. For clarity this will be referred to as the Phi mode. A couple of departures from other Tengwar modes are in the application of the *r* rule[9], and the general suggestion that *tehta* not be placed below *tengwa*. The first is a minor departure; the makeup of Phi words is very regular so the *r* rule is modified for this situation. See below for specifics.

The second is a larger departure. Words in Phi often contain two consecutive vowels. The first *tehta* is placed above the consonant that precedes the vowel, as in the Quenya mode. But to avoid the use of a vowel carrier, the second *tehta* is placed below the same consonant. The letters would then be read as consonant, upper vowel, then lower vowel.

[5] `https://en.wikipedia.org/wiki/Tengwar`

[6] `http://freetengwar.sourceforge.net/tengtelc.html`

[7] `http://freetengwar.sourceforge.net/keylayouts.html`

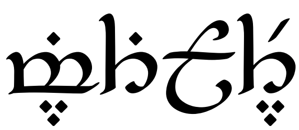
[8] `https://scripts.sil.org/cms/scripts/page.php?site_id=projects&item_id=graphite_fonts`

[9] `https://www.wikihow.com/Write-Tengwar#step-id-16`

This *tehtar* strategy and the use of a single Tengwar character for consonant doubles such as ‹th› or ‹sh› cause Phi script to be very compact and efficient. To illustrate, the word for “love” in Phi is *lothea*. In the Tengwar script it looks like this:

This six-letter word in latin characters requires only two *tengwa* to represent it. Many words in Phi are two to three letters long. This occurs mainly in parts of speech such as particles, prepositions, or pronouns. These words are almost always a single *tengwa* with diacriticals in Tengwar script. All of this results in a very concise writing system.

This is the phrase “I love you” in its simplest vernacular form in Phi:



That is a beautiful sentiment rendered in a beautiful script. For more formal communication in Phi, particles are used to mark parts of speech in order to reduce ambiguity. As well, word markers are often used in Phi Tengwar to distinguish multisyllabic words for clarity. The Phi Tengwar punctuation is outlined in Table 8. As an example, the phrase “I love you” would be written formally as:

In the Phi consonant mapping illustrated in Table 6 there are two *tengwa* that represent the [s] sound: “ϭ” and “9.” Which of the two is used in a word is at the writer’s discretion. Most often it depends on which diacritical marks, if any, are attached to the character. If the ‹s› *tengwa* looks uncomfortable in one orientation, then the alternate might be more effective.

As well, there are two *tengwa* to represent /r/: “ɣ” and “ɲ.” In this case, however, there is a specific usage for both forms. In Phi the initial /r/ in a word is always a trilled [r] and is represented by “ɣ”. Within a word /r/ is always a tapped [ɾ] and is represented by “ɲ”.

__Table 6:__ Consonant grapheme to tengwa mapping

Grapheme	Tengwa	Keypress
‹p›		p
‹t›		t
‹m›		m
‹n›		n
‹r›		r
‹r›		shift-r
‹ph›		shift-p
‹wh›		shift-k
‹th›		shift-t
‹s›		s
‹s›		shift-s
‹sh›		shift-c
‹h›		h
‹w›		w
‹l›		l

Table 7: Vowel grapheme to tehta mapping

Grapheme	Tehta	Keypress
‹i›		i / option-i[10]
‹u›		u / option-u
‹e›		e / option-e
‹o›		o / option-o
‹a›		a / option-a

Table 8: Punctuation to tengwa mapping

Punctuation	Tengwar	Keypress
space	·	,
period	:	shift-,

[10]Keypresses for the lower diacriticals are given for macOS. Use *ctrl-alt* on Windows instead of *option*.

5 Examples

Following are several examples of Phi phrases in formal mode. They are given in romanized text and English, followed by the Tengwar script.

si phiathe soa mia na thanie te phi – My name is Daniel.

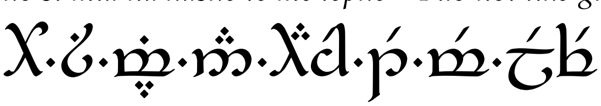
si mia na thi te lothea – I love you.

si whetho na mipho te phi – The sky is blue.

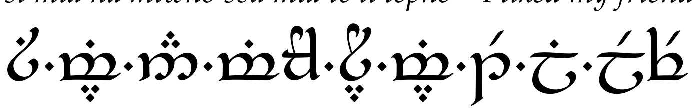
si lo sha na noshea te lephe – They like food.

si mia na sha te lephe – I like it.

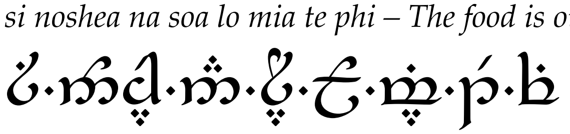
ho si mia na hashe te me lephe – I do not like green!



si mia na miwho soa mia te li lephe – I liked my friend.

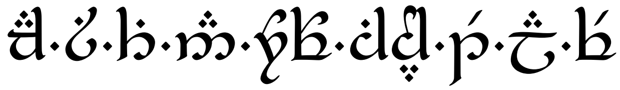


si noshea na soa lo mia te phi – The food is ours.



si noshea soa lo mia na shishoa te phi – Our food is good.

wha si thi na ropho shishoa te la phe – Are you in good health?



si lo mia na lo lipha sathe te lothea – We love the orange trees.